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Department of Computer Engineering

Academic Year 2025–2026

LITERATURE SURVEY

Government Scheme Eligibility and Application Tracking System

A Role-Based Web Application for Digital Management of Government Scheme Applications

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LITERATURE SURVEY

1.Introduction

1.1 Background

The Government of India has introduced numerous welfare schemes over the past several decades to uplift economically weaker sections of society. Schemes such as PM Awas Yojana, PM Kisan Samman Nidhi, and various scholarship programs collectively serve millions of citizens. However, the administration and management of applications for these schemes has traditionally relied on manual, paper-based processes, leading to inefficiencies, delays, and lack of transparency.

The rapid expansion of internet infrastructure and the digital literacy of Indian citizens has created a significant opportunity to digitize the lifecycle of government scheme applications. This project proposes a role-based web application to manage the complete workflow of scheme applications, from submission by applicants to verification by officers and administration by government officials.

1.2 Motivation

The National e-Governance Plan (NeGP) launched by the Government of India emphasizes making all government services accessible to citizens electronically. Despite the existence of portals like DigiLocker, UMANG, and the MyGov platform, a large gap remains in the implementation of structured, workflow-driven application management systems at the scheme level. This project is motivated by the need to bridge that gap by providing a transparent, role-separated, and trackable system for scheme applications.

2.Problem Statement

Existing government portals function primarily as application submission platforms. Once a citizen submits an application, there is limited visibility into the verification process, no structured assignment of officers to applications, and no formal mechanism for communicating rejection reasons or requesting additional documents. The key problems identified are:

- Lack of real-time status tracking for applicants after submission.
- No role-based workflow separating responsibilities of applicants, officers, and administrators.
- No structured mechanism for officers to add remarks or rejection reasons.
- Manual assignment of applications to officers, leading to delays and inconsistency.
- Absence of a centralized dashboard for administrators to monitor overall system activity.
- No digital document verification support linked to the application workflow.

> **Core Objective:** To design and implement a role-based web application that digitizes the complete lifecycle of government scheme applications — from submission and document upload to officer verification and final approval or rejection — with full transparency and real-time tracking at every stage.

3.Review of Existing Systems and Related Work

3.1 E-Governance Framework for Citizen Services — Singh & Sharma (2019)

Singh, R. and Sharma, P. (2019) conducted a comprehensive study on e-governance frameworks deployed across Indian states. The paper analyzed the architecture of several state-level citizen service portals and identified the primary bottleneck as the lack of backend workflow management. The authors noted that most portals focused on data collection rather than data processing. Their study proposed a three-tier architecture consisting of a citizen-facing portal, a government backend, and a verification layer. This work directly influenced the three-role design of our system — Applicant, Officer, and Admin — and confirmed the need for structured workflow management rather than simple form submission.

Limitation identified: The proposed framework was theoretical and did not include implementation details for document verification or real-time status tracking.

3.2 Online Citizen Service Portals: A Comparative Analysis — Rao & Mehta (2020)

Rao, A. K. and Mehta, S. (2020) conducted a comparative analysis of citizen service portals in five countries including India, Estonia, Singapore, South Korea, and Denmark. Their study evaluated portals on parameters such as usability, transparency, security, and efficiency. The findings highlighted that Indian portals scored low on transparency and real-time feedback mechanisms. Estonian and Singaporean portals were identified as best practices due to their complete digitization of application workflows and automated status notifications. This paper provided strong motivation for implementing real-time status tracking and officer remark features in the proposed system.

Limitation identified: The study was observational in nature and did not propose a technical framework for Indian government portals.

3.3 Role-Based Access Control in Web Applications — Ferraiolo et al. (2021)

Ferraiolo, D., Sandhu, R., and Gavrila, S. (2021) reviewed the evolution of Role-Based Access Control (RBAC) models in enterprise and government web applications. The paper presented an updated RBAC framework tailored for multi-stakeholder systems where different users require distinct permissions and views. The authors demonstrated that RBAC significantly reduces the risk of unauthorized data access and simplifies user management. Their framework proposed that roles should be defined at the database level and enforced at the application layer. This research directly informed the design of our system's role management, where each role — Applicant, Officer, Admin — has specific and limited access to system functions.

Limitation identified: The paper focused on access control theory and did not address application workflow or document management in the context of government services.

3.4 Workflow Automation in Public Service Delivery — Kumar & Joshi (2022)

Kumar, V. and Joshi, A. (2022) explored the application of workflow automation techniques in public service delivery systems in developing nations. The study analyzed case studies from state welfare departments in Rajasthan and Maharashtra, where manual processing of benefit applications caused average delays of 45-60 days. The authors implemented an automated

workflow prototype using Java Spring Boot and demonstrated a 68% reduction in processing time. The study concluded that structured workflow management — including automatic routing of applications to available officers, deadline tracking, and escalation mechanisms — is critical for effective e-governance. Their technical stack choice of Java Spring Boot for backend development influenced our technology selection.

Limitation identified: The prototype was deployed in a limited pilot environment and did not scale to real-world government infrastructure. Security features such as JWT authentication were not implemented.

3.5 Digital Document Verification in Government Applications — Patil & Desai (2023)

Patil, N. and Desai, R. (2023) examined the challenges of document verification in digitized government application systems. The paper surveyed 12 government portals and found that 78% lacked a structured document submission and tracking mechanism linked to the application workflow. The authors proposed a document management module integrated into the application workflow, where each uploaded document is linked to both the application and the applicant, and officers can flag specific documents for re-upload or rejection. The proposed module used a relational database schema where documents are stored as file paths with metadata, which closely aligns with our document entity design using file_path and document_name columns in the documents table.

Limitation identified: The proposed system did not implement officer remarks at the document level and lacked a complete frontend interface for document management.

4.Comparison of Existing Systems

The table below summarizes the key features of existing related systems compared to the proposed system:

Feature	PM Awas Portal	NSP Scholarship	DigiLocker	Proposed System
Online Application	✓ Yes	✓ Yes	✗ No	✓ Yes
Real-Time Status Tracking	Limited	Limited	✗ No	✓ Full tracking
Role-Based Access (Admin / Officer / Applicant)	✗ No	Partial	✗ No	✓ Full RBAC
Officer Assignment by Admin	✗ No	✗ No	✗ No	✓ Yes
Officer Remarks / Rejection Reason	✗ No	Partial	✗ No	✓ Full remarks
Document Upload & Tracking	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Admin Dashboard with Analytics	✗ No	✗ No	✗ No	✓ Yes
Workflow-Driven Processing	✗ No	✗ No	✗ No	✓ Yes

5.Technologies Used

5.1 Backend — Java and Spring Boot

Java is a robust, platform-independent, object-oriented programming language widely used for enterprise and government application development. Spring Boot is an extension of the Spring Framework that simplifies the setup and development of production-ready applications. It provides auto-configuration, embedded server support (Apache Tomcat), and a rich ecosystem of libraries for database access (JPA/Hibernate), security (Spring Security), and REST API development.

For this project, Spring Boot 4.0.3 is used to build RESTful APIs that serve data to the frontend. Spring Data JPA is used to interact with the MySQL database through entity classes and repository interfaces, eliminating the need for manual SQL query writing. Spring Security provides role-based access control, ensuring that each user can only access endpoints relevant to their role.

Component	Technology	Purpose
Backend Framework	Spring Boot 4.0.3	REST API development and application logic
ORM Layer	Hibernate / JPA	Database entity mapping and queries
Security	Spring Security + JWT	Authentication and role-based authorization
Build Tool	Apache Maven	Dependency management and build lifecycle
Language	Java 17	Backend business logic implementation

Table 2: Backend Technology Stack

5.2 Database — MySQL

MySQL 8.0 is an open-source relational database management system known for its reliability, performance, and ease of use. It uses Structured Query Language (SQL) for data definition and

manipulation. For this project, MySQL is used to store all persistent data including user accounts, scheme details, applications, documents, and officer remarks.

The database is designed with six normalized tables: roles, users, schemes, applications, documents, and application_remarks. Foreign key constraints ensure referential integrity between related tables. The use of ENUM data types for status fields (PENDING, APPROVED, REJECTED) and boolean is_active columns ensures data consistency and allows soft deletion of records.

5.3 Frontend — HTML, CSS, JavaScript

The frontend of the application is developed using core web technologies — HTML5 for page structure, CSS3 for styling and responsive layout, and vanilla JavaScript for interactivity. This approach was chosen for its simplicity, broad browser compatibility, and ease of integration with Spring Boot's static resource serving capability. Chart.js is used for rendering statistical charts on the admin and officer dashboards. The frontend communicates with the backend via RESTful API calls using the Fetch API in JavaScript.

6.Proposed System

6.1 System Overview

The proposed Government Scheme Eligibility Application and Tracking System is a role-based web application that digitizes the complete lifecycle of government scheme applications. The system consists of three user roles — Applicant, Officer, and Administrator — each with distinct responsibilities and access controls. The application workflow proceeds as follows: an applicant registers, browses available schemes, and submits an application with required documents. The administrator assigns the application to a verification officer. The officer reviews the application and uploaded documents, adds remarks, and approves or rejects the application. The applicant can track the status of their application in real-time at every stage.

6.2 Key Features

- **Role-based access control** — distinct dashboards and permissions for Applicant, Officer, and Admin.
- **Scheme management** — Admin can create, update, and deactivate government schemes.
- **Application workflow** — structured lifecycle from submission to final decision.
- **Document management** — applicants upload documents; officers review them within the application.
- **Officer assignment** — Admin assigns pending applications to available officers.
- **Remarks system** — Officers add remarks and rejection reasons linked to each application.
- **Real-time tracking** — Applicants track their application status with officer remarks visible.
- **Admin analytics dashboard** — charts and statistics for overall system monitoring.

7.Gap Analysis

Based on the review of existing systems and related research papers, the following gaps have been identified which the proposed system aims to address:

Gap Identified in Existing Systems	How Proposed System Addresses It
No structured workflow between submission and decision	Implements a defined workflow: Submitted → Officer Assigned → Under Review → Approved/Rejected
No role separation between admin, officer, and applicant	Full Role-Based Access Control with Spring Security; separate dashboards per role
Applicants cannot track status after submission	Dedicated tracking module with progress indicator and officer remarks visible to applicant
No officer assignment mechanism by admin	Admin dashboard includes dedicated officer assignment section for pending applications
No formal remarks or rejection reason system	Officers can add multiple remarks per application; all remarks visible to applicant in timeline view
No centralized monitoring for administrators	Admin dashboard with analytics charts, stats cards, and full application management

Table 3: Gap Analysis — Existing vs Proposed System

8.Conclusion

The literature survey reveals that while significant work has been done in the area of e-governance and digital citizen services, a clear gap exists in the availability of structured, workflow-driven, role-based application management systems for government welfare schemes in India. Existing portals lack real-time tracking, structured officer assignment, and formal remark mechanisms. The proposed system directly addresses these gaps by implementing a complete application lifecycle management solution with role-based access control, officer workflow, document management, and a transparent tracking interface for applicants. The use of Java Spring Boot, MySQL, and standard web technologies ensures scalability, security, and ease of maintenance.

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